

Fundamentals of Graph Theory

Graph theory is a core topic in discrete mathematics that studies graphs, which are mathematical structures modeling pairwise relations between objects, consisting of vertices (nodes) connected by edges. **Graphs** are essential for analyzing networks, relationships, and optimization problems in computer science and mathematics.

Basic Terminologies

- ✓ **Degree of a Vertex** – The degree of a vertex is the number of edges connected to that vertex.
- ✓ **Path** – sequence of vertices connected by edges
- ✓ **Cycle** – path that starts and ends at the same vertex
- ✓ **Connected Graph** – all vertices are reachable

Graph Types

- **Graph.** A graph is simply a collection of Vertices represented as V , and also known as points or objects, and Edges, represented as E which are connections between those points

We usually write for the notation of graph as: $G = (V, E)$

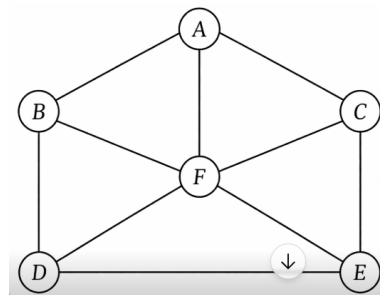


Figure 1. sample graph with vertices and edges

From figure 1, the identified Vertices are A, B, C, D, E, F; and Edges are connections like (A-B), (A-C), (A-F), (F-D), (F-E) etc.

This means that each letter is a vertex and each line connecting them is an edge.

- **Undirected Graph:** Edges are bidirectional, connecting two vertices symmetrically. In an undirected graph, edges do not have a specific direction, meaning connections go both ways. If two places are connected, you can travel in either direction.
- **Directed Graph (Digraph):** Edges have a specific direction from an origin to a destination. Edges in a digraph have a direction, i.e., the edges have arrows indicating the direction of traversal.

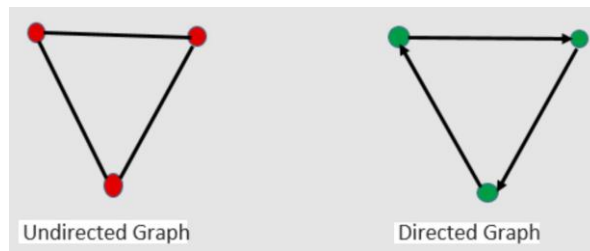


Figure 2. undirected and directed graphs

- **Simple Graph:** Contains no loops (edges connecting a vertex to itself) or multiple edges between the same pair of vertices. It does not contain more than one edge between the pair of vertices. A simple railway track connecting different cities is an example of a simple graph.

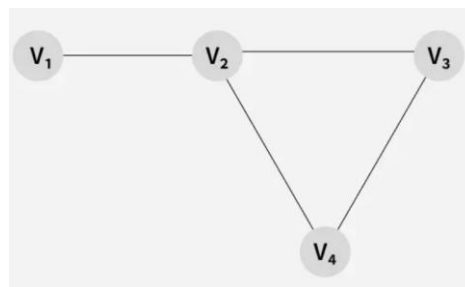


Figure 3. a simple graph

- **Multigraph.** Any graph which contains some parallel edges but doesn't contain any self-loop is called a multigraph. For example a Road Map.
 - **Parallel Edges.** If two vertices are connected with more than one edge then such edges are called parallel edges that many routes but one destination.
 - **Loop.** An edge of a graph that starts from a vertex and ends at the same vertex is called a loop or a self-loop.

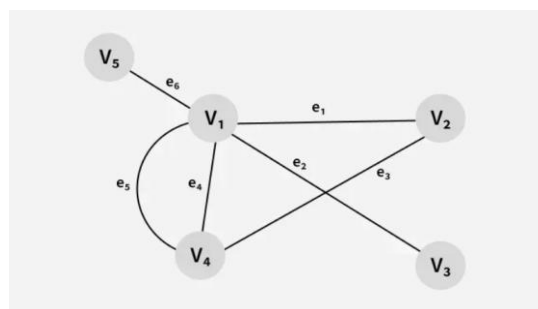


Figure 4. graph with parallel edges and with a loop

- **Null Graph.** A graph of order n and size zero is a graph where there are only isolated vertices with no edges connecting any pair of vertices. A null graph is a graph with no edges. In other words, it is a graph with only vertices and no connections between them. A null graph can also be referred to as an edgeless graph, an isolated graph, or a discrete graph.

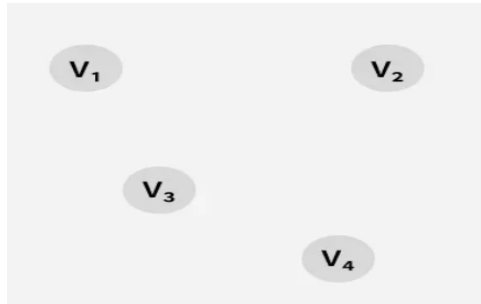


Figure 5. An example of a null graph

- **Complete Graph.** A simple graph with n vertices is called a complete graph if the degree of each vertex is $n-1$, that is, one vertex is attached with $n-1$ edges or the rest of the vertices in the graph. A complete graph is also called Full Graph.

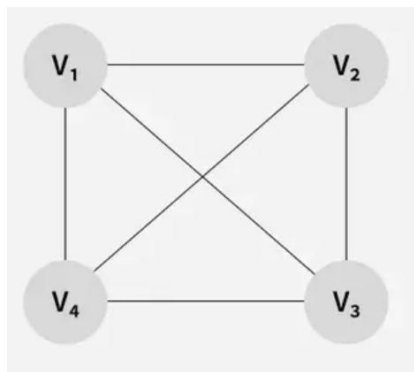


Figure 6. Complete Graph

- **Weighted Graphs.** A weighted graph is a graph where each edge has a number (weight) that represents distance, cost, or time. These graphs help find the shortest or cheapest paths. Examples include Google Maps, airline routes, and delivery networks.

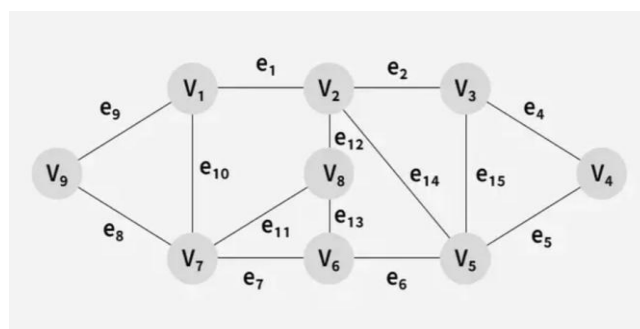


Figure 7. Weighted graph

- **Unweighted Graphs.** An unweighted graph is a graph where all edges are treated equally, with no extra values like distance or cost. It simply shows connections between points. Examples include basic social networks and metro maps without travel times.

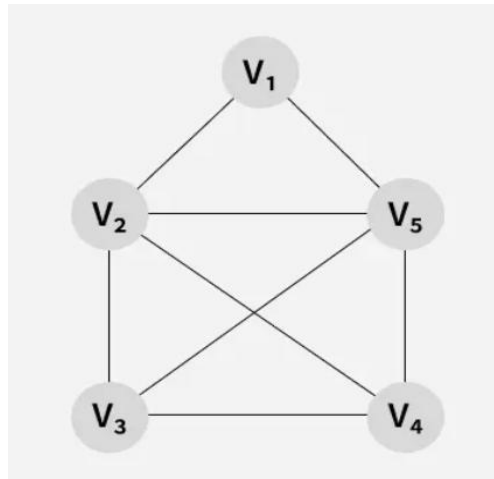


Figure 8. Unweighted graph

- **Pseudo Graph.** A pseudograph is a type of graph that allows for the existence of self-loops (edges that connect a vertex to itself) and multiple edges (more than one edge connecting two vertices). In contrast, a simple graph is a graph that does not allow for loops or multiple edges.

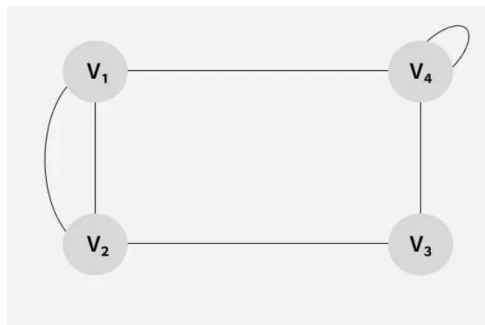


Figure 9. Pseudograph

- **Regular Graph.** A regular graph is a type of undirected graph in which every vertex has the same number of edges (or neighbors). In other words, all vertices in a regular graph have the same degree.

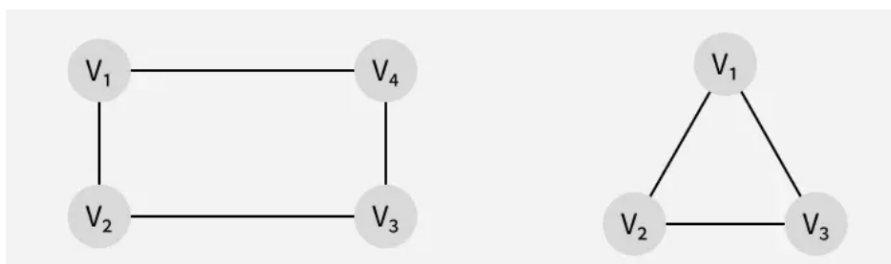


Figure 10. Regular Graph

- **Trees.** A tree is a connected graph that contains no cycles. In other words, there is exactly one path between any two vertices.

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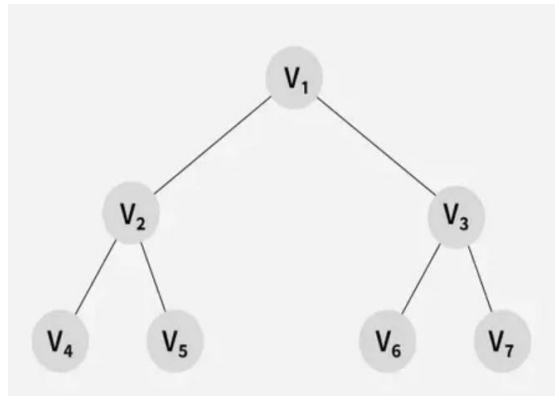


Figure 10. A Tree

<https://www.geeksforgeeks.org/dsa/graph-types-and-applications/>